PROBLEM STATEMENT 03 · QDRANT EDGE

# FieldMind

Memory for field technicians that works with no network, knows what it is allowed to share, and syncs with the fleet whenever a link exists.



**Team Compilers** · github.com/yabhinav1/fieldmind

# The most useful field knowledge is created where the network is worst

1

## No signal

Plant floors, basements and remote sites drop the connection for hours. The work does not stop.

2

## Mixed notes

One note can hold a fault the whole fleet needs and a colleague's name that must never leave the device.

3

## Drift

Devices that work apart come back with different versions of the truth, and someone's edit gets lost.

FieldMind is one process on the device: Qdrant Edge for memory, small on-device models for meaning, names, reranking, speech and photos, SQLite for what is waiting, and a dashboard a technician can use with gloves on. Two devices run 24/7 on arm64 servers against Qdrant Cloud; the same code runs on a laptop or a Raspberry Pi.

# The device that keeps working

1

## Record

Write, dictate or photograph what you saw. Speech is transcribed on the device by Whisper; the device shows what it will share before you save, and learns from your overrides.

2

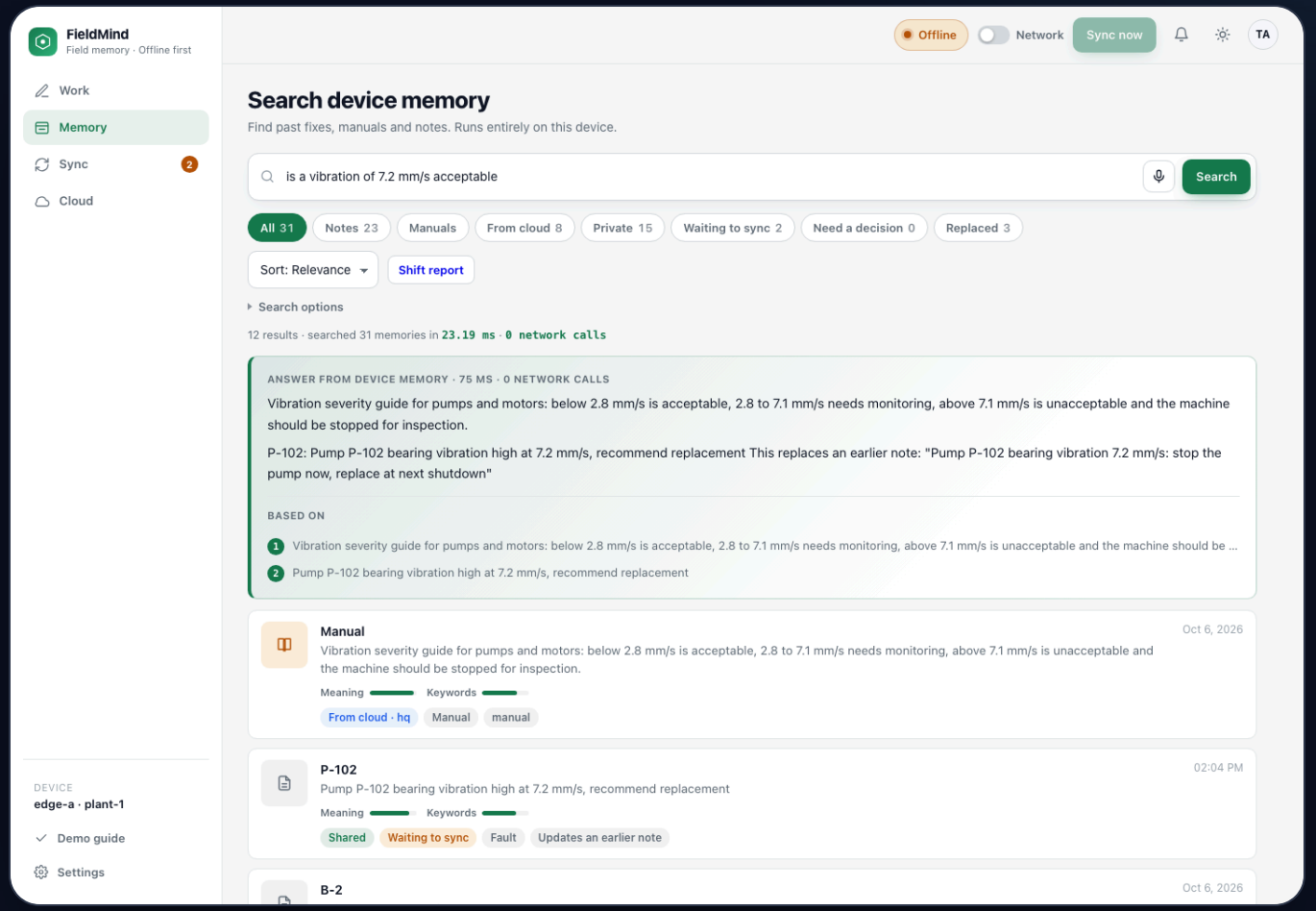
## Find

Hybrid search over your notes, your photos and headquarters manuals in about a millisecond. A cross-encoder rereads the best candidates.

3

## Ask

A cited answer written on the device. Every number, asset and verdict is checked against the note it cites. No cloud call.



# One device process, one cloud collection

## EDGE DEVICE

**Local shard**
Written here, private too. Sealed on disk with the device key.

**Replica shard**
Cloud knowledge, or what nearby devices shared when the cloud was gone.

**Policy engine**
Decides what may leave. Remembers the person's overrides as vectors, never text.

**Embeddings**
bge-small (or multilingual) and BM25; MiniLM reranker; CLIP for photos.

**Journal**
Outbox, conflicts, sealed activity. Bounded, survives restarts.

**Answer model**
llama3.2 3B, optional, with a source check on every answer.

**Push** →
compare and swap, payload-only when the text is unchanged

**Pull** ←
diff, or one snapshot when far behind

## QDRANT SERVER

Shared notes
Masked notes
Headquarters manuals
Photo thumbnails

Other devices sync here too, and with each other over the local network when it is unreachable.

# The device decides what leaves

| WHAT THE TECHNICIAN WROTE | WHAT HAPPENS |
| --- | --- |
| Pump P-102 bearing vibration high at 7.2 mm/s | Shared |
| Smell of gas near compressor K-4 | Shared · Urgent, sent first |
| Suresh and Priya replaced the coupling, call 9876543210 | Shared, names and number masked |
| Technician Ravi Kumar reported chest pain | Private, stays on the device |
| SCADA panel login password is … | Private, cannot be overridden |
| Operator slipped on oil near P-102, no injury | Shared · Flagged: reads as two kinds |

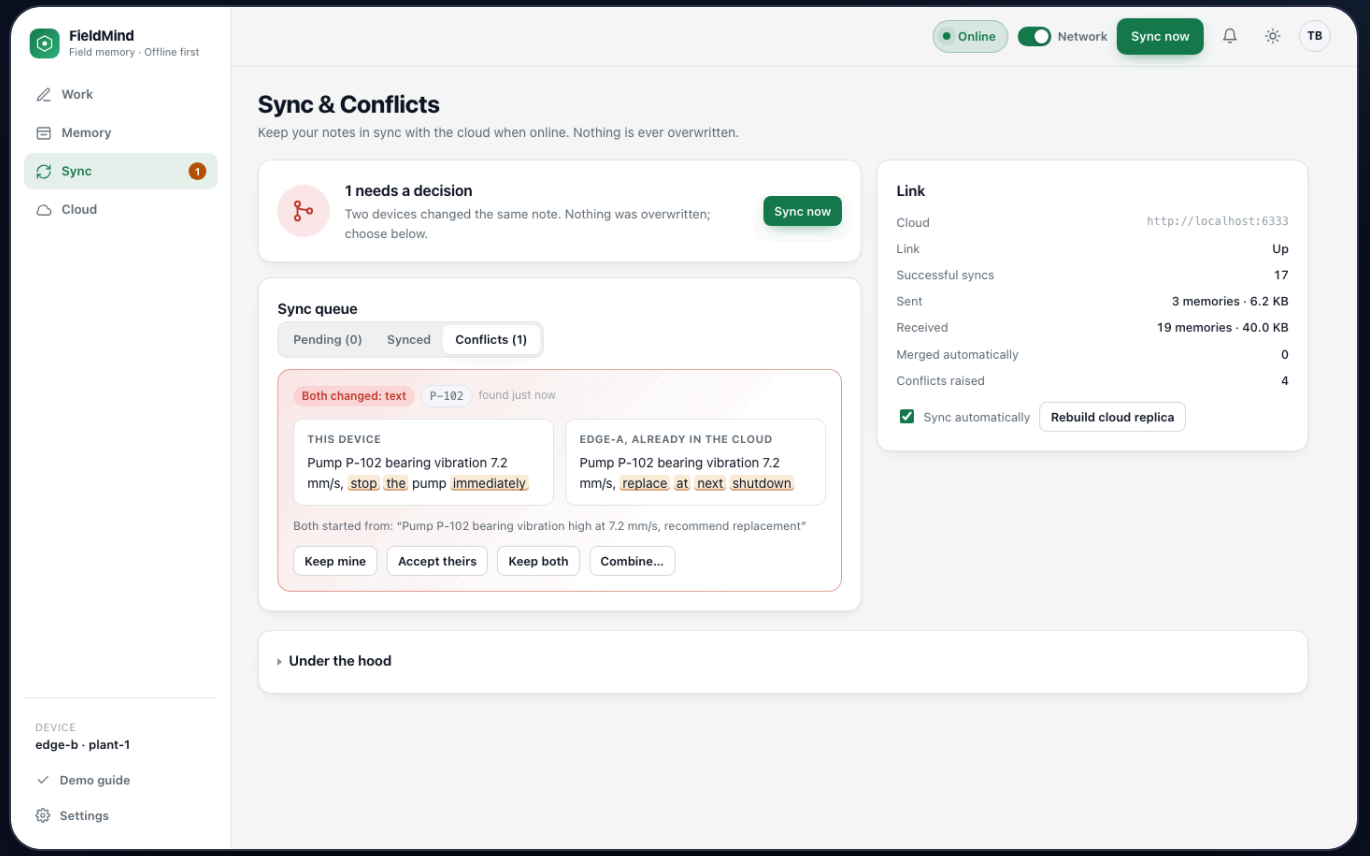
About 20 ms on the device: a name recognition model, pattern detectors for credentials, contacts, identity numbers, vehicles, badges and addresses, and a semantic classifier. Vectors for a masked note are computed from the masked text. Override a decision and notes like it follow your choice from then on.

# When two devices disagree

1. **Nothing is overwritten**
   An upload lands only if the cloud is still at the revision the device last saw, confirmed by a per-write id. An edit made mid-upload is rebased, not lost.

2. **Most edits merge**
   Changes to different fields combine automatically, field by field.

3. **A person settles the rest**
   Keep mine, accept theirs, keep both or combine. Even while offline. One click stages a conflict for the demo.



# What Qdrant does for us

| QDRANT FEATURE | HOW FIELDMIND USES IT |
| --- | --- |
| Edge shards, in process | Three per device: local notes, a cloud replica, and photos with a CLIP vector each |
| Built-in BM25 with IDF | Keyword half of hybrid search, with no server |
| Dense vectors and filters | Meaning search narrowed by asset, scope and status; photos found by what they show |
| Shard snapshots | A device far behind restores its replica from one file |
| Conditional upsert and payload overwrite | Compare and swap, so devices never clobber each other; payload-only patches skip the vectors |
| One schema on both sides | The same dense and sparse vectors on edge and server, versioned payloads |

qdrant-edge-py 0.8.0 with Qdrant Server 1.19.2, pinned and tested in CI.

# Measured, not estimated

**1 ms**
hybrid search across both shards

**0**
network calls to search, answer, or find a photo

**20 ms**
to decide what may leave the device

**1–3 s**
for a cited answer from a 3B model

**106**
automated tests on every push: two devices, a real Qdrant Server, a real browser

**30**
checks in the live end-to-end run

Ryzen 7 7435HS laptop, RTX 3050 4 GB. Also verified on Linux through a real network disconnect, and in production on two arm64 servers with Qdrant Cloud. Dictation: about 0.4 s for a six-second clip on a CPU.

# Five moments in the demo

1. Switch the network off and keep working
2. Ask a question, get a cited answer
3. Watch names get masked before upload
4. Reconnect: urgent notes go first
5. Two devices disagree, a person decides

### ALSO IN THE BOX

Dictation transcribed on the device (Whisper) · Private text sealed on disk · First PIN claimed with a console code · Devices learn from each other when the cloud is gone · Photos searchable by what they show · Encrypted backup to a replacement device · Shift report · One-click staged conflict · Installable on a phone · Hindi and other languages optional · /healthz and /metrics

### NEXT

TLS for remote dashboards · Faces and name plates in photos · Photos between peers

# Try it live

LIVE DEVICE

**fieldmindedgea.hidenplay.net**

edge-a, running 24/7 on an arm64 server with Qdrant Cloud. Ask us for the PIN, then click **Demo guide**. A second device, edge-b, syncs with it.

CODE AND TESTS

**github.com/yabhinav1/fieldmind**

MIT. 106 tests on every push against a real Qdrant Server and a real browser. `docker compose --profile demo up` runs the whole thing on any laptop.

FIVE THINGS TO TRY IN TWO MINUTES

Turn the Network switch off and keep working · Click Dictate and speak a note · Ask “is a vibration of 7.2 mm/s acceptable” · Type a note with a name and a phone number and watch the preview mask them · Demo guide → Stage a conflict, then decide